Each side label is a SQUARED vector length

Orthogonal triangles in observation space; schematic, not to scale.

A GENERAL NESTED MODELS

$$SSE_R = (SSE_R - SSE_F) + SSE_F$$

B REDUCED MODEL: INTERCEPT ONLY

$$SST = SSR + SSE$$

The GREEN side joins the two fitted vectors; its squared length is the extra SS.